Certified Analytical Bounds for Multi-Phase Hypersonic Reachable Sets

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Abstract

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The vehicle is a rigid body, not a point mass (Appendix A). Its state

$$x = \big(r, \theta, \phi,\ u, v, w,\ q_0, q_1, q_2, q_3,\ \omega_1, \omega_2, \omega_3\big) \in \mathcal{X} \subseteq \mathbb{R}^{13}, \tag{2.1}$$

augmented by mass, recession depth s and modal state $(\eta, \dot{\eta})$ where the corresponding mode or coupling is active, carries geocentric position, body-frame velocity, attitude quaternion and body rates as components. The attitude coordinates are redundant: every admissible state satisfies

$$q_0^2 + q_1^2 + q_2^2 + q_3^2 = 1, \tag{2.2}$$

so the thirteen coordinates carry twelve degrees of freedom. The control

$$c = \big(\vec{\delta},\ \vec{\tau}_{\mathrm{rcs}},\ \varsigma\big) \in U, \tag{2.3}$$

collects flap deflections $\vec{\delta}$, reaction-control moments $\vec{\tau}_{\mathrm{rcs}}$ and, in a propulsive mode, throttle ς; $U \subset \mathbb{R}^{n_c}$ is compact.

list deflection, RCS, and throttle limits (one citation each). compactness follows.

Crucially the incidence angles (α, β) are *outputs* of the attitude solution (**??**), not commanded quantities: trim is not assumed solvable, and attitudes the vehicle cannot hold are excluded automatically rather than by fiat. The reachable set cannot contain a state reached only by an unattainable incidence.

elaborate

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zhao2020optimal asm. 2 (p2063): the field has to be transverse to every guard. that constrains where the $\bar{q}$ threshold can sit (a trajectory grazing the surface tangentially breaks it, prove transversality for the corridor? restrict the threshold placement like altitude band?).

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the converse comes with a hypothesis attached. zhao2020optimal asm. 11 (p2066): need $F(t,x,U)$ compact *and convex* or measures satisfying Liouville need not come from trajectories. cor. 13 (p2066) gives control-affine dynamics with convex U as sufficient (flap deflections enter nonlinearly? check.)

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A Six-Degree-of-Freedom Equations of Motion

This appendix records the rigid-body dynamics the body of the paper treats as given. Nothing here is a contribution; the derivation is standard entry mechanics assembled once with a fixed set of frames and a fixed parametrization. Frame errors are the classic 6-DOF defect and are invisible in the assembled algebra, so the transformations are stated explicitly and used without exception. cite sources

A.1 State vector and frames

Frames. Four frames are used throughout.

- $\mathcal{I}$: Earth-centred inertial (ECI), non-rotating.
- $\mathcal{E}$: Earth-centred Earth-fixed (ECEF), rotating with the constant planetary rate $\vec{\omega}_E = \omega_E\,\hat{z}_{\mathcal{I}}$.
- $\mathcal{L}$: local geocentric frame at the vehicle, with unit vectors $(\hat{e}_r, \hat{e}_\theta, \hat{e}_\phi)$ pointing radially outward, east, and north.
- $\mathcal{B}$: body frame, fixed at the instantaneous centre of mass, axes along the vehicle's principal design directions.

The attitude quaternion $\vec{q}$ carries $\mathcal{I} \to \mathcal{B}$; the geocentric angles (θ, ϕ) carry $\mathcal{I} \to \mathcal{L}$.

The direction-cosine matrix built from $\vec{q}$,

$$\mathbf{R}_{B/I}(\vec{q}) = \begin{pmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_1q_2+q_0q_3) & 2(q_1q_3-q_0q_2) \\ 2(q_1q_2-q_0q_3) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_2q_3+q_0q_1) \\ 2(q_1q_3+q_0q_2) & 2(q_2q_3-q_0q_1) & q_0^2-q_1^2-q_2^2+q_3^2 \end{pmatrix}, \tag{A.1}$$

is polynomial of degree two in $\vec{q}$, which is the property Appendix C exploits. The geocentric map is

$$\mathbf{R}_{L/I}(\theta, \phi) = \begin{pmatrix} \cos\phi\cos\theta & \cos\phi\sin\theta & \sin\phi \\ -\sin\theta & \cos\theta & 0 \\ -\sin\phi\cos\theta & -\sin\phi\sin\theta & \cos\phi \end{pmatrix}. \tag{A.2}$$

State vector. Before augmentation the state carries thirteen components,

$$x = \big(\underbrace{r, \theta, \phi}_{\text{position}},\ \underbrace{u, v, w}_{\text{body velocity}},\ \underbrace{q_0, q_1, q_2, q_3}_{\text{attitude}},\ \underbrace{p, q, r_b}_{\text{body rates}}\big) \in \mathbb{R}^{13}, \tag{A.3}$$

augmented by mass m when a propulsive mode is present (Appendix A.6), and by the recession depth s and modal state $(\eta, \dot{\eta})$ once the aerothermoelastic loop is closed (Appendix A.3 and Section 2.5). Here (r, θ, ϕ) are geocentric radius, longitude and latitude; (u, v, w) are the body-axis components of the *planet-relative* velocity

$$\vec{V} \;=\; \vec{V}_{\mathcal{I}} - \vec{\omega}_E \times \vec{r} - \vec{w}_{\text{wind}}, \tag{A.4}$$

with $\vec{V}_{\mathcal{I}}$ the inertial velocity and $\vec{w}_{\text{wind}}$ the local wind (Appendix A.5); and (p, q, r_b) are the body components of the inertial angular velocity $\vec{\omega} \equiv \vec{\omega}_{B/I}$.

A.2 Translational equations

Newton's law holds in $\mathcal{I}$: $m\,\dot{\vec{V}}_{\mathcal{I}}\big|_{\mathcal{I}} = \vec{F}$, with $\vec{F} = \vec{F}_g + \vec{F}_a + \vec{F}_t$ (gravity, aerodynamic, thrust). We want the rate of the *relative* velocity resolved in $\mathcal{B}$. Writing $\vec{V}_{\mathcal{I}} = \vec{V} + \vec{\omega}_E \times \vec{r}$ and differentiating in $\mathcal{I}$,

$$\dot{\vec{V}}_{\mathcal{I}}\big|_{\mathcal{I}} = \dot{\vec{V}}\big|_{\mathcal{I}} + \vec{\omega}_E \times \vec{V}_{\mathcal{I}} = \underbrace{\dot{\vec{V}}\big|_{\mathcal{B}} + \vec{\omega} \times \vec{V}}_{\text{transport of } \vec{V}} + \vec{\omega}_E \times \vec{V} + \vec{\omega}_E \times (\vec{\omega}_E \times \vec{r}), \tag{A.5}$$

where the transport theorem $\frac{d}{dt}\big|_{\mathcal{I}} = \frac{d}{dt}\big|_{\mathcal{B}} + \vec{\omega}\times$ was applied to $\vec{V}$. Solving for the body-frame rate,

$$\boxed{\dot{\vec{V}}\big|_{\mathcal{B}} = \frac{1}{m}\big(\vec{F}_a + \vec{F}_t\big) + \vec{g} - \big(\vec{\omega} + \vec{\omega}_E\big) \times \vec{V} - \vec{\omega}_E \times (\vec{\omega}_E \times \vec{r})} \tag{A.6}$$

with $\vec{F}_g = m\vec{g}$. The term $(\vec{\omega} + \vec{\omega}_E) \times \vec{V}$ collects body-rotation transport and Coriolis; expanding $\vec{\omega} = \vec{\omega}_{B/E} + \vec{\omega}_E$ recovers the familiar $\vec{\omega}_{B/E} \times \vec{V} + 2\,\vec{\omega}_E \times \vec{V}$ split, the second being the Coriolis acceleration proper.

A.3 Rotational equations

Angular momentum about the center of mass is $\vec{h} = \mathbf{I}\vec{\omega}$, and $\dot{\vec{h}}\big|_{\mathcal{I}} = \vec{M}$; transporting to $\mathcal{B}$,

$$\boxed{\mathbf{I}\,\dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I}\vec{\omega}) - \dot{\mathbf{I}}\,\vec{\omega}} \tag{A.7}$$

with $\vec{M} = \vec{M}_a + \vec{M}_{\text{rcs}}$. The gyroscopic term is quadratic in $\vec{\omega}$, hence polynomial once $\mathbf{I}$ is; the inversion $\dot{\vec{\omega}} = \mathbf{I}^{-1}(\cdots)$ is carried as the linear solve $\mathbf{I}\dot{\vec{\omega}} = (\cdots)$ so that no rational function of the state enters the embedding.

Remark A.1 (The inertia tensor is not constant). Ablation removes mass from the nose and structural deformation redistributes it, so $\mathbf{I} = \mathbf{I}(s, \eta)$ through the recession depth s and modal amplitudes η (Section 2.5), and Euler's equations acquire the $\dot{\mathbf{I}}\vec{\omega}$ term in Eq. (A.7). We carry it rather than drop it: the whole point of going to 6-DOF is to keep the deformation→incidence→moment→attitude loop closed, and that loop runs through $\mathbf{I}(s, \eta)$. Its magnitude is bounded by the recession and modal-amplitude envelopes, so carrying it costs one Jacobian block, not a new uncertainty.

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